

Day 9

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[3]: df = pd.read_csv("Housing.csv")

*[4]: df.head()
df.tail()
df.shape
df.columns

[4]: Index(['price', 'area', 'bedrooms', 'bathrooms', 'stories', 'mainroad',
'guestroom', 'basement', 'hotwaterheating', 'airconditioning',
'parking', 'prefarea', 'furnishingstatus'],
dtype='object')

*[5]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  ------                -
0   price                 545 non-null   int64
1   area                  545 non-null   int64
2   bedrooms              545 non-null   int64
3   bathrooms              545 non-null   int64
4   stories                545 non-null   int64
5   mainroad               545 non-null   object
6   guestroom              545 non-null   object
7   basement               545 non-null   object
8   hotwaterheating        545 non-null   object
9   airconditioning        545 non-null   object
10  parking                545 non-null   int64
11  prefarea               545 non-null   object
12  furnishingstatus       545 non-null   object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB

[6]: df.describe()
```

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[6]:
```

	price	area	bedrooms	bathrooms	stories	parking
count	5.450000e+02	545.000000	545.000000	545.000000	545.000000	545.000000
mean	4.766729e+06	5150.541284	2.965138	1.286239	1.805505	0.693578
std	1.870440e+06	2170.141023	0.738064	0.502470	0.867492	0.861586
min	1.750000e+06	1650.000000	1.000000	1.000000	1.000000	0.000000
25%	3.430000e+06	3600.000000	2.000000	1.000000	1.000000	0.000000
50%	4.340000e+06	4600.000000	3.000000	1.000000	2.000000	0.000000
75%	5.740000e+06	6360.000000	3.000000	2.000000	2.000000	1.000000
max	1.330000e+07	16200.000000	6.000000	4.000000	4.000000	3.000000

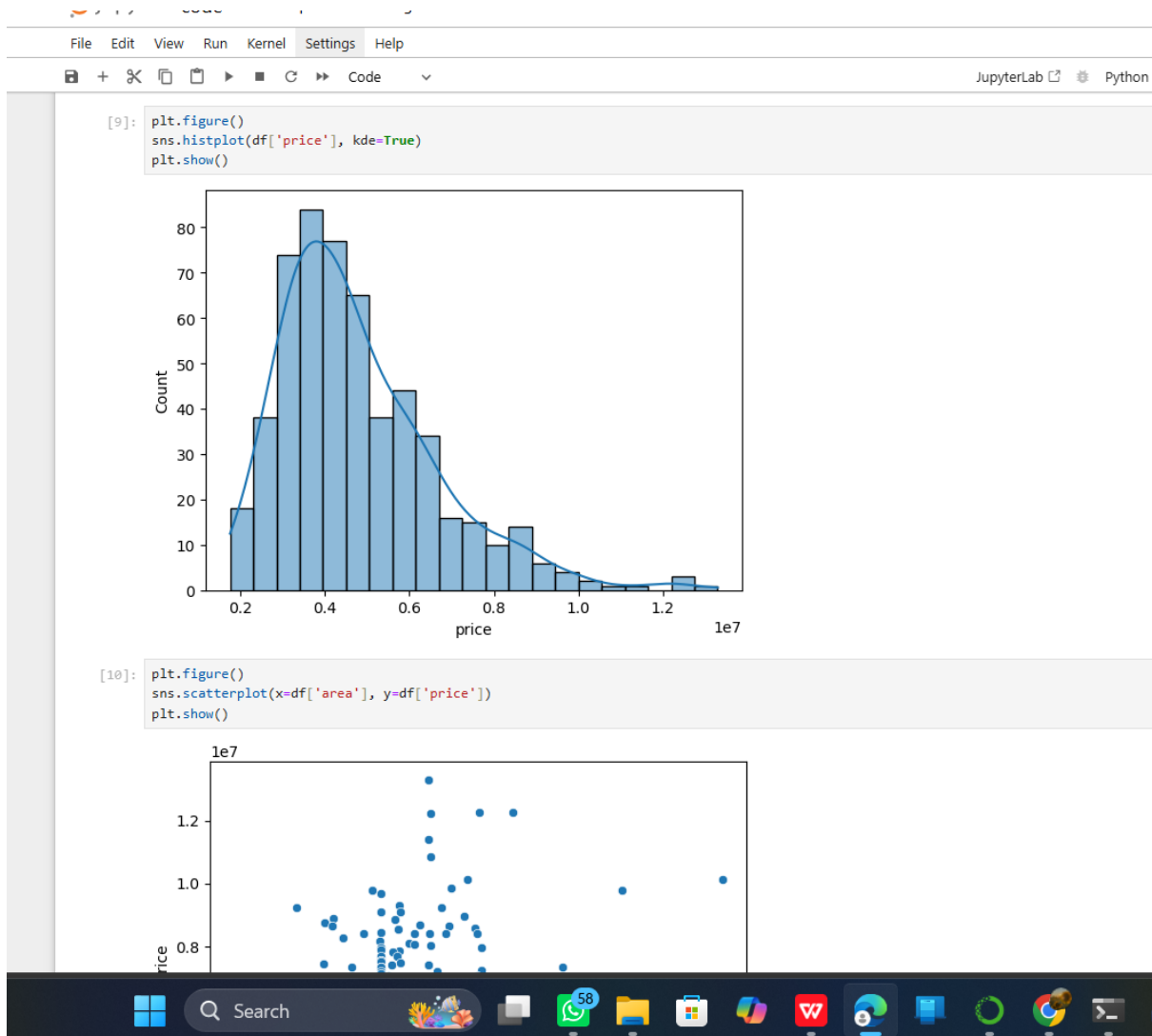
```
[7]: df.duplicated().sum()
```

```
[7]: 0
```

```
[8]: corr = df.corr(numeric_only=True)
```

```
plt.figure()
sns.heatmap(corr, annot=True)
plt.show()
```





area

```
[11]: df.select_dtypes(include='object')
```

```
[11]:
```

	mainroad	guestroom	basement	hotwaterheating	airconditioning	prefarea	furnishingstatus
0	yes	no	no	no	yes	yes	furnished
1	yes	no	no	no	yes	no	furnished
2	yes	no	yes	no	no	yes	semi-furnished
3	yes	no	yes	no	yes	yes	furnished
4	yes	yes	yes	no	yes	no	furnished
...	...	...	...	...	...	...	...
540	yes	no	yes	no	no	no	unfurnished
541	no	no	no	no	no	no	semi-furnished
542	yes	no	no	no	no	no	unfurnished
543	no	no	no	no	no	no	furnished
544	yes	no	no	no	no	no	unfurnished

545 rows × 7 columns

```
[12]: df['mainroad'].value_counts()
```

```
[12]:
```

mainroad	count
yes	468
no	77

Name: count, dtype: int64

```
[13]: df.isnull().sum()
```

```
[13]:
```

price	0
area	0
bedrooms	0
bathrooms	0
stories	0
mainroad	0
guestroom	0
basement	0
hotwaterheating	0
airconditioning	0

542	yes	no	no	no	no	no	unfurnished
543	no	no	no	no	no	no	furnished
544	yes	no	no	no	no	no	unfurnished

545 rows × 7 columns

```
[12]: df['mainroad'].value_counts()
```

```
[12]: mainroad
yes    468
no      77
Name: count, dtype: int64
```

```
[13]: df.isnull().sum()
```

```
[13]: price          0
area            0
bedrooms        0
bathrooms       0
stories         0
mainroad        0
guestroom       0
basement        0
hotwaterheating 0
airconditioning 0
parking         0
prefarea        0
furnishingstatus 0
dtype: int64
```

```
[14]: df.fillna(df.mean(numeric_only=True), inplace=True)
```

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[17]: df.drop_duplicates(inplace=True)
```

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[ ]:
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